Yuanhang Zhang

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Research Interests

- **LLM-driven autonomous research**
  - Designing LLM agents to autonomously explore, model and optimize scientific problems.
- **AI for Physics**
  - Quantum systems: Developing general purpose "large quantum models" for quantum simulation and computation.
  - Dynamical systems: Applying graph neural networks to model, predict, and optimize complex physical processes.
- **Physics for AI**
  - Harnessing long-range order in nonlinear dynamical systems to develop efficient computational paradigms.
  - Leveraging neuromorphic devices based on novel materials to enhance the energy efficiency of machine learning.

Education

- Ph.D. in Physics, University of California San Diego — Jan 2020 - Aug 2024
  Thesis: The physics of computing with memory — Advisor: Prof. Massimiliano Di Ventra
- B.Sc. Physics with minor in Computer Science, University of Science and Technology of China
  Sep 2015 - Jul 2019

Experiences

- Postdoctoral Researcher, UC San Diego — Sep 2024 – Now
  Advisors: Prof. Ivan Schuller, Prof. Massimiliano Di Ventra
- Teaching assistant, Electricity & Magnetism Lab, UC San Diego — Jan 2020 – Mar 2020
- Postbaccalaureate Researcher, Tsinghua University — Aug 2019 – Dec 2019
  Advisor: Prof. Dong-Ling Deng

Publications

[1] **Zhang, Y. H.**, Sipling, C., Qiu, E., Schuller, I. K., & Di Ventra, M. (2024). Collective dynamics and long-range order in thermal neuristor networks. ***Nature Communications****, 15(1), 6986.*

[2] **Zhang, Y. H.**, Zheng, P. L., Zhang, Y., & Deng, D. L. (2020). Topological quantum compiling with reinforcement learning. ***Physical Review Letters****, 125(17), 170501.*

[3] **Zhang, Y. H.**, Sipling, C., & Di Ventra, M. (2025). Memory-induced long-range order drag. *arXiv preprint arXiv:2510.24712.*

[4] Palin, V., Hofer, J. A., Ghazikhanian, N., **Zhang, Y. H.**, & Schuller, I. K. (2025). Impact of substrate thermal conductivity on VO_2 resistive-switching devices. *Physical Review Applied*, 24(4), 044071.

[5] Du, Y., Zhu, Y., **Zhang, Y. H.**, Hsieh, M.H., Rebentrost, P., Gao, W., Wu, Y.D., Eisert, J., Chiribella, G., Tao, D. & Sanders, B.C. (2025). Artificial intelligence for representing and characterizing quantum systems. *arXiv preprint arXiv:2509.04923.*

[6] Sipling, C., **Zhang, Y. H.**, & Di Ventra, M. (2025). Memory-induced long-range order in dynamical systems. *Physical Review E, 112(1), 014124.*

[7] Nguyen, D. C., Chetaille, T., **Zhang, Y. H.**, Pershin, Y. V., & Di Ventra, M. (2025). On the solvable-unsolvable transition due to noise-induced chaos in digital memcomputing. *arXiv preprint arXiv:2506.14928*.

[8] **Zhang, Y. H.**, Jia, Z., Wu, Y. C., & Guo, G. C. (2025). An efficient algorithmic way to construct Boltzmann machine representations for arbitrary stabilizer code. *Entropy*, *27*(6), 627.

[9] Sipling, C., **Zhang, Y. H.**, & Di Ventra, M. (2025). Phase-Space Engineering and Dynamical Long-Range Order in Memcomputing. *arXiv preprint arXiv:2506.10149*.

[10] **Zhang, Y. H.**, & Di Ventra, M. (2025). A Generative Neural Annealer for Black-Box Combinatorial Optimization. *arXiv preprint arXiv:2505.09742*.

[11] **Zhang, Y. H.**, & Di Ventra, M. (2025). Implementation of digital memcomputing using standard electronic components. *International Journal of Circuit Theory and Applications*, *53*(4), 2447-2465.

[12] Sun, J., Sipling, C., **Zhang, Y. H.**, & Di Ventra, M. (2024). Memory in neural activity: long-range order without criticality. *arXiv preprint arXiv:2409.16394.*

[13] Qiu, E., **Zhang, Y. H.**, Di Ventra, M., & Schuller, I. K. (2023). Reconfigurable cascaded thermal neuristors for neuromorphic computing. *Advanced Materials, 2306818.*

[14] Primosch, D., **Zhang, Y. H.**, & Di Ventra, M. (2023). Self-averaging of digital memcomputing machines. *Physical Review E, 108(3), 034306.*

[15] Nguyen, D.C., **Zhang, Y. H.**, Di Ventra, M. & Pershin, Y.V. (2023). Hardware implementation of digital memcomputing on small-size FPGAs. *2023 IEEE 66th International Midwest Symposium on Circuits and Systems.*

[16] **Zhang, Y. H.**, & Di Ventra, M. (2023). Transformer quantum state: A multipurpose model for quantum many-body problems. *Physical Review B, 107(7), 075147.*

[17] **Zhang, Y. H.**, & Di Ventra, M. (2022). Efficient quantum state tomography with mode-assisted training. *Physical Review A, 106(4), 042420.*

[18] **Zhang, Y. H.**, & Di Ventra, M. (2021). Directed percolation and numerical stability of simulations of digital memcomputing machines. *Chaos: An Interdisciplinary Journal of Nonlinear Science, 31(6), 063127.*

[19] Zhao, J., **Zhang, Y. H.**, Shao, C. P., Wu, Y. C., Guo, G. C., & Guo, G. P. (2019). Building quantum neural networks based on a swap test. *Physical Review A, 100(1), 012334.*

[20] Jia, Z. A., **Zhang, Y. H.**, Wu, Y. C., Kong, L., Guo, G. C., & Guo, G. P. (2019). Efficient machine-learning representations of a surface code with boundaries, defects, domain walls, and twists. *Physical Review A, 99(1), 012307.*

Selected Talks

[1] Long-range order and its relevance in machine learning, APS Global Physics Summit 2025

[2] Computing with long-range order: from AI to combinatorial optimization, invited seminar at USTC, 2024

[3] Collective dynamics and memory-induced long-range order in spiking oscillator arrays, APS March Meeting 2024 and featured student talk at APS Far West Section 2023

[4] Neuromorphic computing with thermal interactions, invited talk at NIST, 2023

[5] Transformer Quantum State: A Multi-Purpose Model for Quantum Many-Body Problems, APS March Meeting 2023

[6] Self-Averaging of Digital MemComputing Machines, APS March Meeting 2023

[7] Towards a general-purpose model for quantum many-body problems, seminar at Tsinghua University, 2022

[8] Quantum State Tomography with Mode-assisted Training, APS March Meeting 2022

[9] Topological Quantum Compiling with Reinforcement Learning, seminar at University of Oxford, 2021